

Chapter 35. Effects of Mechanical Ventilation on Control of Breathing

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Effects of Mechanical Ventilation on Control of Breathing: Introduction

The main reasons for instituting mechanical ventilation are to decrease the work of breathing, support gas exchange, and buy time for other interventions to reverse the cause of respiratory failure.¹ Mechanical ventilation can be applied in patients who are making or not making respiratory efforts, whereby assisted or controlled modes of support are used, respectively.¹ In patients without respiratory efforts, the respiratory system represents a passive structure, and thus the ventilator is the only system that controls breathing. During assisted modes of ventilator support, the patient's system of control of breathing is under the influence of the ventilator pump.^{2–4} In the latter instance, ventilatory output is the final expression of the interaction between the ventilator and the patient's system of control of breathing. Thus, physicians who deal with ventilated patients should know the effects of mechanical ventilation on control of breathing, as well as their interaction. Ignorance of these issues may prevent the ventilator from achieving its goals and also lead to significant patient harm.

Physiology

The respiratory control system consists of a motor arm, which executes the act of breathing, a control center located in the medulla, and a number of mechanisms that convey information to the control center.^{5,6} Based on information, the control center activates spinal motor neurons that subserve the respiratory muscles (inspiratory and expiratory); the intensity and rate of activity vary substantially between breaths and between individuals. The activity of spinal motor neurons is conveyed, via peripheral nerves, to respiratory muscles, which contract and generate pressure (P_{mus}). According to equation of motion, P_{mus} at time t during a breath is dissipated in overcoming the resistance (Rrs) and elastance (Ers) of the respiratory system (inertia is assumed to be negligible) as follows:

$$P_{mus}(t) = Rrs \times \dot{V}(t) + Ers \times \Delta V(t) \quad (1)$$

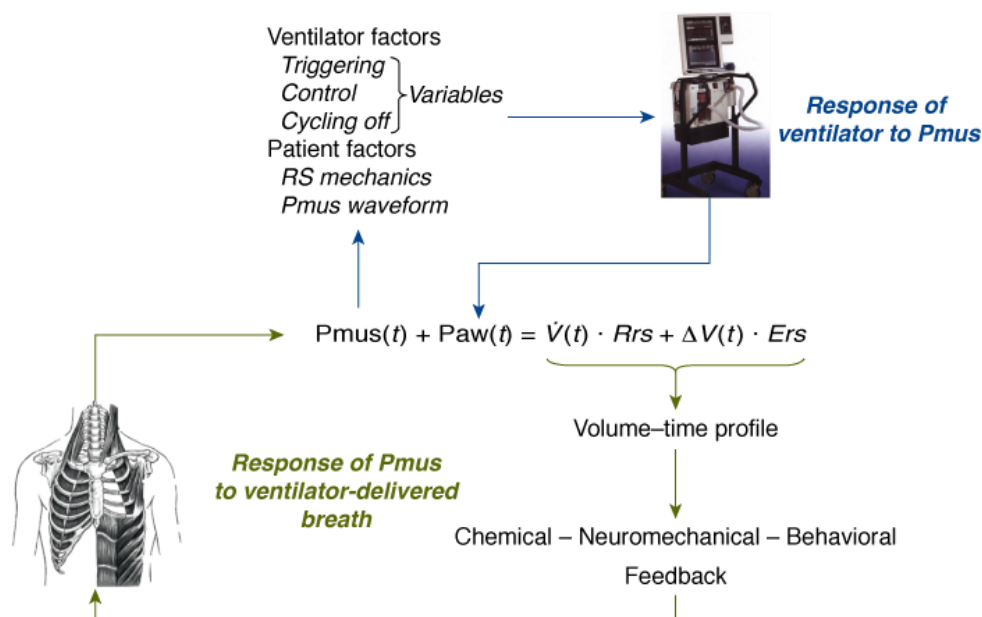
where $\Delta V(t)$ is instantaneous volume relative to passive functional residual capacity and $\dot{V}(t)$ is instantaneous flow. Equation (1) determines the volume–time profile and, depending on the frequency of respiratory muscle activation, ventilation. Volume–time profile affects P_{mus} via neuromechanical feedback; inputs generated from other sources (cortical inputs) may modify the function of control center. Ventilation, gas-exchange properties of the lung, and cardiac function determine arterial blood gases, termed *arterial oxygen tension* (Pa_{O_2}) and *arterial carbon dioxide tension* (Pa_{CO_2}), which, in turn, affect the activity of control center via peripheral and central chemoreceptors (chemical feedback). This system can be influenced at any level by diseases or therapeutic interventions.

During mechanical ventilation, the pressure provided by the ventilator (Paw) is incorporated into the system.³ Thus, the total pressure applied to respiratory system at time t [$P_{TOT}(t)$] is the sum of $P_{mus}(t)$ and $Paw(t)$. As a result, the equation of motion is modified as follows:

$$\begin{aligned} P_{TOT}(t) &= P_{mus}(t) + Paw(t) \\ &= \dot{V}(t) \times Rrs + \Delta V(t) \times Ers \end{aligned} \quad (2)$$

The relationships of Equation (2) determine the volume–time profile during mechanical ventilation, which via neuromechanical, chemical, and behavioral feedback systems affects the P_{mus} waveform (Fig. 35-1). The ventilator pressure, by changing flow and volume, may influence these feedback systems and thus alter either the patient's control of breathing itself or its expression. In addition, P_{mus} , depending on several factors, alters the Paw waveform (Fig. 35-1). Thus, during assisted mechanical ventilation (i.e., $P_{mus} \neq 0$), ventilatory output is not under the exclusive influence of patient's control of breathing; instead, it represents the final expression of an interaction between ventilator-delivered pressure and patient respiratory effort.

Figure 35-1



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Schematic of variables that determine the volume-time profile during mechanical ventilation. Neuromechanical, chemical, and behavioral feedback systems are the main determinants of Pmus. The functional operation of the ventilator mode (triggering, control, and cycling-off variables) and patient-related factors (namely, respiratory system mechanics and the Pmus waveform) determine the response of the ventilator to Pmus. $\Delta V(t)$, instantaneous volume relative to passive functional residual capacity of respiratory system; E_{rs} , elastance of the respiratory system; $P_{aw}(t)$, airway (ventilator) pressure; $P_{mus}(t)$, instantaneous respiratory muscle pressure; R_{rs} , resistance of the respiratory system; RS , respiratory system; $\dot{V}(t)$, instantaneous flow.

Effects of Mechanical Ventilation on Feedback Systems

Chemical Feedback

Chemical feedback refers to the response of Pmus to Pa_{O_2} , Pa_{CO_2} , and pH.⁵⁻⁷ In spontaneously breathing and mechanically ventilated patients, this system is an important determinant of respiratory motor output both during wakefulness and sleep.⁷⁻¹¹

Mechanical ventilation can influence chemical feedback simply by altering the three variables Pa_{O_2} , Pa_{CO_2} , and pH. Hypoxemia, hypercapnia, or acidemia may be corrected by mechanical ventilation and thus modify activity of the medullary respiratory controller via peripheral and central chemoreceptors.^{5,12} The effects of mechanical ventilation on gas-exchange properties of the lung are beyond the scope of this chapter and are discussed in [Chapter 37](#). In this chapter, the fundamental elements of the response of respiratory motor output to chemical stimuli, their relationship to unstable breathing, and the operation of chemical feedback during mechanical ventilation are reviewed.

Response of Respiratory Motor Output to Chemical Stimuli

Carbon Dioxide Stimulus

Carbon dioxide (CO_2) is a powerful stimulus of breathing.^{5,12} This stimulus, expressed by Pa_{CO_2} , largely depends on the product of tidal volume (V_T) and breathing frequency (f) (i.e., minute ventilation) according to Equation (3):

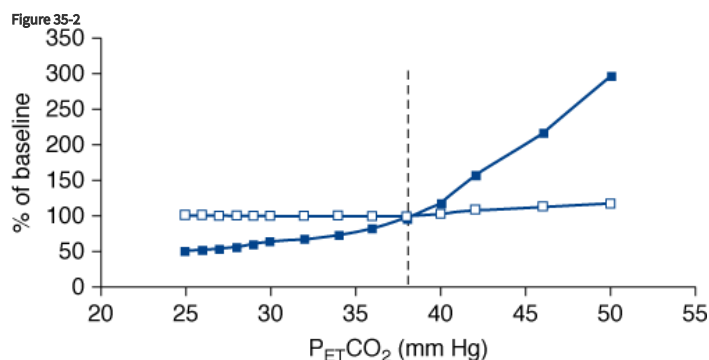
$$Pa_{CO_2} = 0.863 \frac{\dot{V}_{CO_2}}{V_T \times f(1 - V_D/V_T)} \quad (3)$$

where $\dot{V}_{CO_2}$ is CO_2 production, and V_D/V_T is the dead-space-to-tidal-volume ratio. Because minute ventilation is an adjustable variable in ventilated patients, understanding the relationship between respiratory motor output and CO_2 stimuli is of fundamental importance.

Several studies have examined the respiratory motor output to CO_2 in ventilated, conscious, healthy subjects.^{7,13-16} Major findings include

1. Manipulation of Pa_{CO_2} over a wide range has no appreciable effect on respiratory rate. Despite hypocapnia, subjects continue to trigger the ventilator with a rate similar to that of eucapnia. Respiratory rate increases slightly when Pa_{CO_2} approaches values well above eucapnia ([Fig. 35-2](#)).

2. The intensity of respiratory effort (respiratory drive) increases progressively as a function of P_{CO_2} . This response is evident even in hypocapnic range. The response slope increases progressively with increasing CO_2 stimuli, reaching its maximum in the vicinity of eucapnic values (see Fig. 35-2).
3. There is no fundamental difference in the response to CO_2 between various ventilator modes.
4. Above eupnea, the slope of the response does not differ significantly with that observed during spontaneous breathing, suggesting that mechanical ventilation per se does not considerably modify the sensitivity of respiratory system to CO_2 .



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Schematic of the response of respiratory frequency (*open squares*) and pressure-time product of the inspiratory muscles per breath (an index of the intensity of patient effort, *closed squares*), both expressed as a percentage of values during spontaneous eupnea (baseline), to CO_2 challenge in conscious healthy subjects ventilated with a high level of ventilator assistance. $P_{ET}CO_2$ is end-tidal P_{CO_2} , and the *dotted vertical line* is $P_{ET}CO_2$ during spontaneous breathing (eupnea). Contrast the vigorous response of intensity of inspiratory effort to CO_2 , even in the hypocapnic range, with the response of respiratory frequency, which remains at eucapnic level over a broad range of CO_2 stimuli. The response is based on data from references 7 and 13 to 16.

During sleep (or sedation), the response of respiratory motor output to CO_2 differs substantially from that during wakefulness, secondary to loss of the suprapontine neural input to the medullary respiratory controller.^{10,17} In ventilated sleeping subjects, a decrease in Pa_{CO_2} by a few millimeters of mercury causes apnea.¹⁰ Respiratory rhythm is not restored until Pa_{CO_2} has increased significantly above eupneic levels. The difference between eupneic Pa_{CO_2} and Pa_{CO_2} at apneic threshold, referred to as CO_2 reserve,¹⁸ depends on several factors (see [Response of Respiratory Motor Output to Chemical Stimuli—Chemical Stimuli and Unstable Breathing](#)). This reserve determines the propensity of an individual to develop breathing instability during sleep; propensity increases as CO_2 reserve decreases. Similar to wakefulness, the response of respiratory motor output to CO_2 is mediated mainly by the intensity of respiratory effort, whereas respiratory rate decreases abruptly to zero (apnea) when the CO_2 apneic threshold is reached.¹⁹

Other Chemical Stimuli

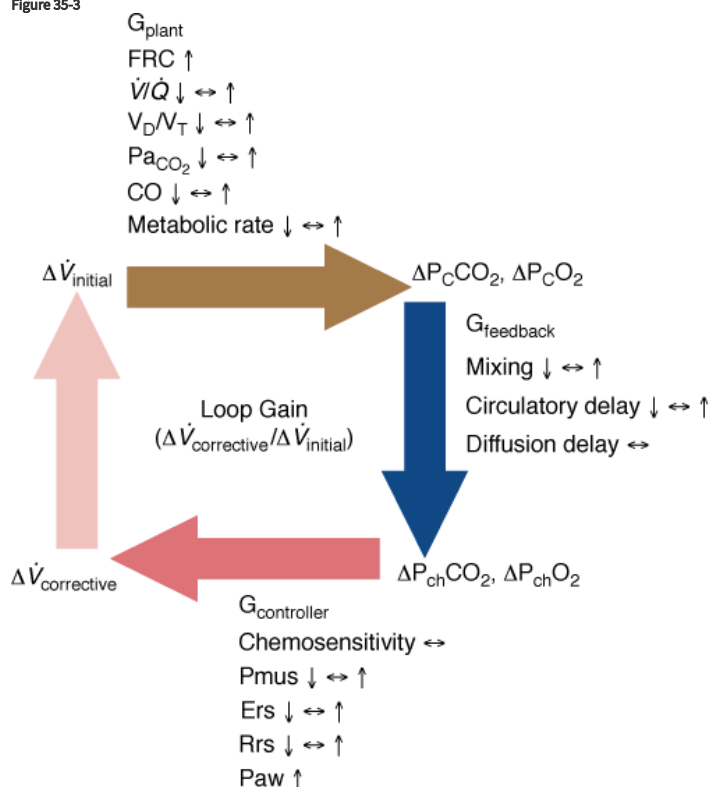
The effects of mechanical ventilation on the response of respiratory motor output to stimuli other than CO_2 have not been studied adequately. In a steady state during wakefulness, the effects of [oxygen](#) (O_2) and pH on breathing pattern are similar qualitatively to that observed with CO_2 : Changes in O_2 and pH mainly alter the intensity of patient effort, whereas respiratory rate is affected considerably less.^{5,12} There is no reason to expect a different response pattern during mechanical ventilation. Indeed, this is the case regarding the hypoxic response in normal conscious subjects ventilated in assist-control mode during eupnea.²⁰ Indirect data also revealed that during eupnea, the sensitivity of respiratory motor output to hypoxia was not modified by mechanical ventilation.²⁰ During mild hypocapnia, however, the response was attenuated, whereas at moderate hypocapnia (end-tidal P_{CO_2} approximately 31 mm Hg) the response was negligible. The latter observations may be relevant clinically because ventilated patients do not always keep Pa_{CO_2} at eucapnic levels and can become hypocapnic.¹⁶

Chemical Stimuli and Unstable Breathing

The response pattern of respiratory motor output to CO_2 during sleep is relevant to the occurrence of periodic breathing in mechanically ventilated patients. Studies indicate that this breathing pattern might increase the morbidity and mortality of critically ill patients because it can cause sleep fragmentation and patient-ventilator dyssynchrony.^{21–23} Sleep deprivation may cause serious cardiorespiratory,^{24,25} neurologic,^{26,27} immunologic, and metabolic consequences.^{28–31}

The following is a brief review of the factors that can lead to unstable breathing. In a closed system governed mainly by chemical control (such as occurs during sleep or sedation), a transient change in ventilation at a given metabolic rate ($\Delta\dot{V}_{\text{initial}}$) will result in a transient change in alveolar gas tensions. This change is sensed by peripheral and central chemoreceptors, which, after a variable delay, exert a corrective ventilatory response ($\Delta\dot{V}_{\text{corrective}}$) that is in the opposite direction to the initial perturbation^{32,33} (Fig. 35-3). The ratio of $\Delta\dot{V}_{\text{corrective}}$ to $\Delta\dot{V}_{\text{initial}}$ defines the loop gain of the system.³² Loop gain is a dimensionless index that is the mathematical product of three types of gains: plant gain (the relationship between the change in gas tensions in mixed pulmonary capillary blood and $\Delta\dot{V}_{\text{initial}}$), feedback gain (the relationship between gas tensions at the chemoreceptor level and those at the mixed pulmonary capillary level), and controller gain (the relationship between $\Delta\dot{V}_{\text{corrective}}$ and the change in gas tensions at the chemoreceptor level) (Fig. 35-3). Loop gain has both a magnitude and a dynamic component.^{32,33} In this system, instability occurs when the corrective response is 180 degrees out of phase with initial disturbance (dynamic component) and loop gain is greater than 1 (magnitude component). This instability leads to fluctuation in chemical stimuli, namely, P_{CO_2} . If P_{CO_2} reaches the apneic threshold, apnea occurs.

Figure 35-3



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Schematic of the variables that determine the propensity of an individual to develop periodic breathing in a closed system dominated by chemical feedback. Loop gain is the product of three gains: plant, feedback, and controller. Instability occurs when $\Delta\dot{V}_{\text{corrective}}$ (the final response) is 180 degrees out of phase with $\Delta\dot{V}_{\text{initial}}$ (the transient initial perturbation) and $\Delta\dot{V}_{\text{corrective}}/\Delta\dot{V}_{\text{initial}}$ is greater than 1. Mechanical ventilation, by affecting almost all variables of the system (↑, increase; ↔, no change; ↓, decrease), may change both the magnitude and the dynamic component of loop gain and thus the propensity of an individual to develop periodic breathing. CO, cardiac output; ΔP_{CCO_2} and ΔP_{CO_2} , the difference in partial pressures of CO_2 and O_2 in mixed pulmonary capillary blood, respectively; ΔP_{chCO_2} and ΔP_{chO_2} , the difference in partial pressure of CO_2 and O_2 at chemoreceptors (peripheral and central), respectively; E_{rs} and R_{rs} , elastance and resistance of respiratory system, respectively; FRC, functional residual capacity; LG, G_{plant} , G_{feedback} , and $G_{\text{controller}}$, loop, plant, feedback, and controller gains, respectively; P_{aCO_2} , alveolar partial pressure of CO_2 ; P_{aw} , airway (ventilator) pressure; P_{mus} , pressure developed by respiratory muscles; $\dot{V}/\dot{Q}$, ventilation-perfusion ratio; $V_{\text{D}}/V_{\text{T}}$, dead-space fraction.

Positive-pressure breathing exerts multiple effects on loop gain by influencing almost all the factors that determine plant, feedback, and controller gains. The effects are complex and at times opposing and variable (Table 35-1; see also Fig. 35-3). Nevertheless, the effect of mechanical ventilation on controller gain exerts the most powerful influence on the propensity to develop breathing instability.^{8,19,21,23} The magnitude and direction of the change in controller gain depends on the ventilator mode, the level of assistance, the mechanics of the respiratory system, and the P_{mus} waveform (see the section [Interactive Effects of Patient-Related Factors and Ventilator on Control of Breathing](#)).^{8,16,19,21} Disease states as well as medications (e.g., sedatives) also may interfere with the effects of mechanical ventilation on loop gain. For example, positive-pressure ventilation may increase or decrease cardiac output, causing corresponding changes in circulatory delay depending on cardiac function and intravascular volume (see Chapter 36).³⁴⁻³⁷ It has been shown that nocturnal mechanical ventilation in patients with congestive heart failure decreases the frequency of Cheyne-Stokes breathing, presumably by causing an increase in cardiac output

secondary to afterload reduction.^{38–40} Sedatives at moderate doses, commonly used in ventilated patients, decrease considerably the loop gain, partly mitigating the effect of mechanical ventilation on controller gain and thus promote ventilatory stability.⁴¹

Table 35-1: Effects of Mechanical Ventilation on Gain Factors and Gain Changes

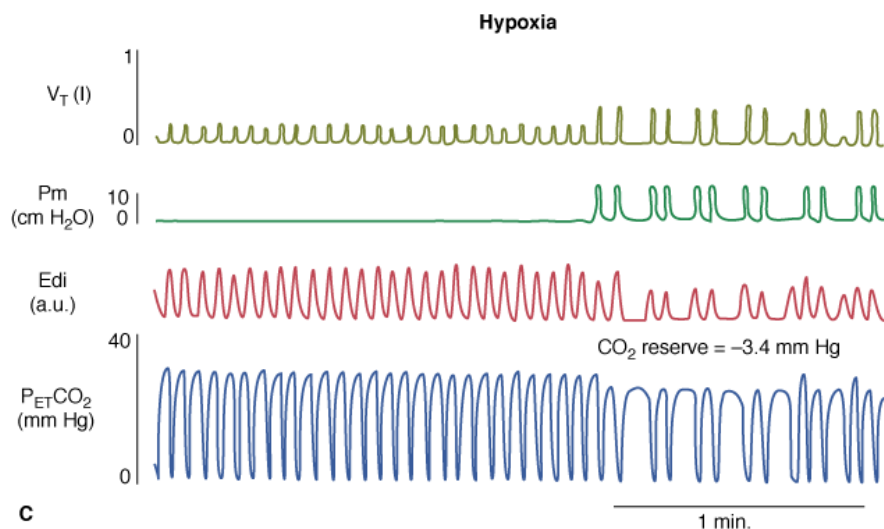
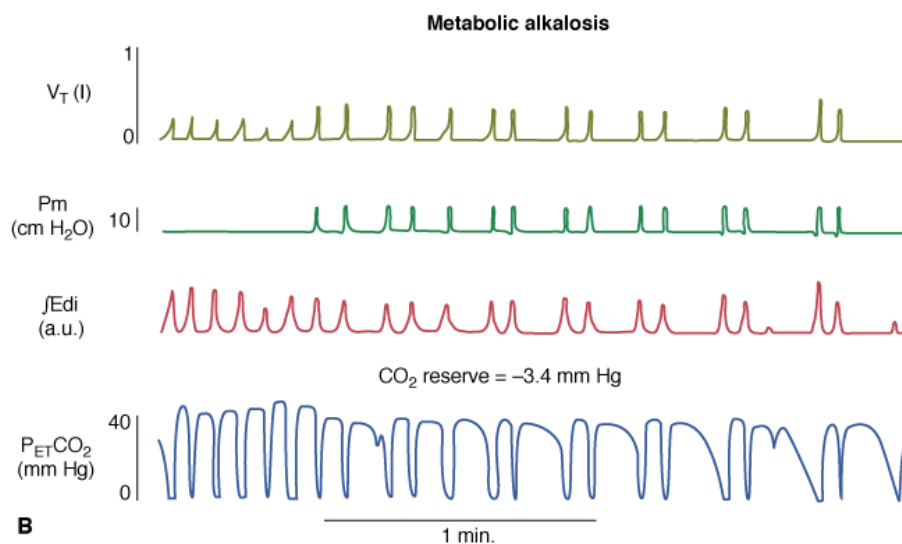
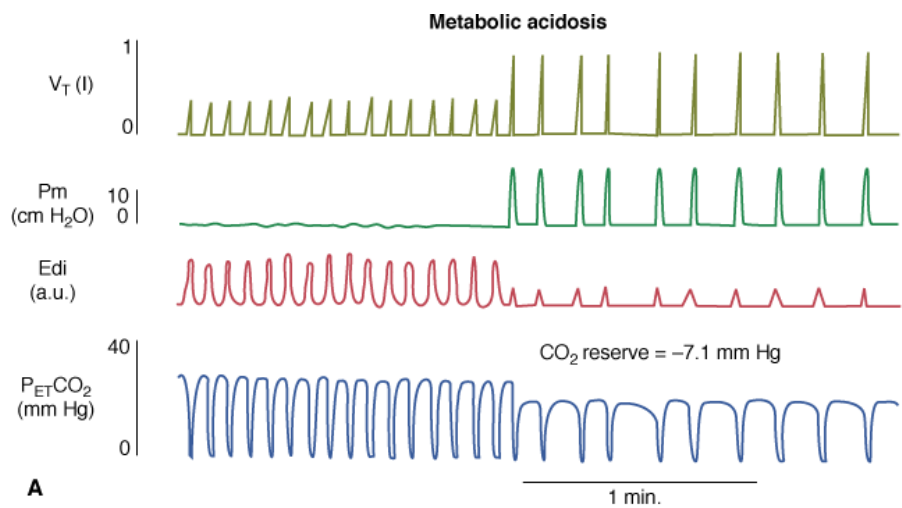
Gain Factors (Influence)	Ventilator Effect *	Gain Change
Lung volume (stabilizing)	↑	↓G _{plant}
Cardiac output (destabilizing)	↓	↑G _{plant} , ↑G _{feedback}
Thoracic blood volume (destabilizing)	↓	↑G _{feedback}
Paw response to P _{mus} (destabilizing)	↑	↑G _{controller}
Alveolar P _{CO₂} (stabilizing)	↓	↓G _{plant}
Alveolar P _{O₂} (stabilizing)	↑	↓G _{plant} , ↓G _{controller}
Respiratory elastance (destabilizing)	↓	↑G _{controller}

Abbreviations: ↓, decrease; ↑, increase; Paw, airway pressure; P_{mus}, respiratory muscle pressure.

*Mechanical ventilation may also exert opposite effects on the various gain factors.

In addition to CO₂, O₂ and pH can play a key role in producing unstable breathing in ventilated patients during sleep (or sedation). It is well known that hypoxia, acting via peripheral chemoreceptor stimulation, decreases Pa_{CO₂}. The result reduces the plant gain (stabilizing influence); for a given change in alveolar ventilation, Pa_{CO₂} will change less when baseline Pa_{CO₂} is low than when it is high.¹⁸ Hypoxia, however, increases the controller gain to a much greater extent⁴² because the slope of ventilatory response to CO₂ below eupnea increases,¹² a highly destabilizing influence.^{32,33} Similar principles apply if pH is considered as a chemical stimulus; acidemia decreases the plant gain (lowers Pa_{CO₂}) and increases, to a much lesser extent, the controller gain.^{18,42} During mechanical ventilation, the propensity to unstable breathing in the face of changing O₂ and pH stimuli depends on a complex interaction between the effects of these stimuli and mechanical ventilation on plant, feedback, and controller gains (Fig. 35-4; see also Table 35-1).

Figure 35-4



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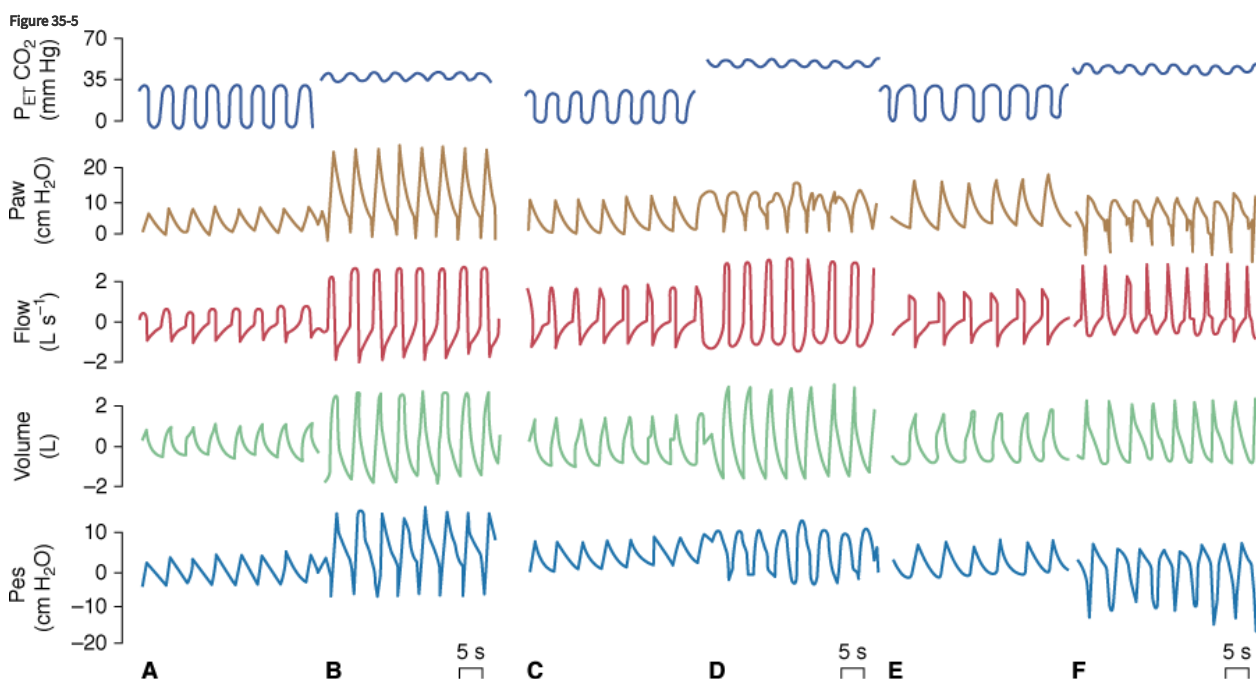
Tidal volume (V_T), airway pressure (P_m), integrated diaphragmatic electrical activity (Edi , arbitrary units), and partial pressure of end-tidal CO_2 (P_{ETCO_2}) in a tracheostomized dog during non-rapid eye movement sleep without and with pressure-support ventilation at a pressure level that caused periodic breathing.

(A) At a background of 5 hours of metabolic acidosis (pH 7.34, HCO_3^- 16 mEq/L, P_{aCO_2} 30 mm Hg). (B) At a background of 1 hour of metabolic alkalosis (pH 7.51, HCO_3^- 35 mEq/L, P_{aCO_2} 44 mm Hg). (C) During hypoxia (P_{aO_2} 47 mm Hg, P_{aCO_2} 31 mm Hg). At a background of metabolic acidosis, CO_2 reserve was quite high; consequently, the pressure level that caused periodic breathing (20 cm H_2O) was significantly higher than the corresponding values (approximately 10 cm H_2O) during metabolic alkalosis or hypoxia. Hyperventilation during spontaneous breathing was similar during metabolic acidosis and hypoxia (similar

stabilization influence via a decrease in plant gain secondary to low P_{aCO_2} , indicating that the destabilizing influence of hypoxia was caused by an increase in controller gain (hypoxic increase in the slope of CO_2 below eupnoea). (Used, with permission, from Dempsey et al. *J Physiol.* 2004;560:1–11, based on data from Nakayama H, Smith CA, Rodman JR, et al. Effect of ventilatory drive on carbon dioxide sensitivity below eupnea during sleep. *Am J Respir Crit Care Med.* 2002;165:1251–1260.)

Operation of Chemical Feedback

The ventilator mode is a major determinant of driving pressure for flow and thus arterial blood gases. Before discussing the operation of chemical feedback, it is useful to review briefly the functional features of three main modes of assisted ventilation, namely, assist-control ventilation (ACV), pressure-support ventilation (PSV), and proportional-assist ventilation (PAV) (for detailed descriptions, see [Chapters 6, 8, and 12](#)). [Figure 35-5](#) shows the response of the ventilator to respiratory effort in a representative subject ventilated with each mode in the presence and absence of CO_2 challenge.¹⁶ With CO_2 challenge, P_{aw} decreases with ACV, it remains constant with PSV, and it increases with PAV. Pressure-time product of inspiratory muscle pressure (PTP-P_{mus}) is an accurate index of the intensity of inspiratory effort.⁴³ With ACV, the ratio of V_T to PTP-P_{mus} per breath (neuroventilatory coupling) decreases with increasing P_{mus}; the ratio is largely independent of inspiratory effort with PAV. With PSV, V_T /PTP-P_{mus} per breath may change in either direction with increasing P_{mus}, depending on factors such as the level of pressure assist and cycling-off criterion, change in P_{mus}, and mechanics of the respiratory system. With PSV, in the absence of active termination of pressure delivery (with expiratory muscle contraction), the ventilator delivers a minimum V_T , which may be quite high, depending on the pressure level, mechanics of the respiratory system, and cycling-off criterion.¹⁹



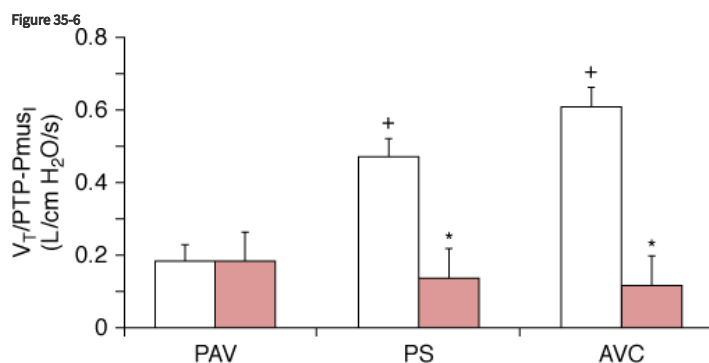
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End-tidal carbon dioxide tension (P_{ETCO_2}), airway pressure (P_{aw}), flow (inspiration up), volume (inspiration up), and esophageal (P_{es}) pressure in a representative subject during proportional-assist ventilation (**A, B**), pressure-support ventilation (**C, D**), and volume-control ventilation (**E, F**) in the absence (**A, C, E**) and presence (**B, D, F**) of CO_2 challenge. With CO_2 challenge, P_{aw} decreases with assist-control ventilation (the ventilator antagonizes patient's effort); it remains constant with pressure-support ventilation (no relationship between patient effort and level of assist); and it increases with proportional-assist ventilation (positive relationship between effort and pressure assist). (Used, with permission from Mitrouska J, Xirouchaki N, Patakas D, et al. Effects of chemical feedback on respiratory motor and ventilatory output during different modes of assisted mechanical ventilation. *Eur Respir J.* 1999;13:873–882.)

Assume that in a ventilated patient P_{aCO_2} drops because of an increase in the set level of assistance or decrease in metabolic rate and/or V_D/V_T ratio.⁴⁴

During wakefulness, patients will react to this drop by decreasing the intensity of their inspiratory effort, whereas the breathing frequency will remain relatively constant (see “[Response of Respiratory Motor Output to Chemical Stimuli](#),” above). The extent to which a patient is able to prevent respiratory alkalosis via operation of chemical feedback depends almost exclusively on the relationship between the intensity of patient inspiratory effort and the volume delivered by the ventilator (i.e., V_T /PTP-P_{mus}). Similarly, if P_{aCO_2} increases (decrease in assistance level, increase in metabolic rate and/or V_D/V_T ratio), the patient will increase the intensity of inspiratory effort and, to much lesser extent, respiratory frequency. Thus, V_T /PTP-P_{mus} per breath is critical for the effectiveness of chemical feedback to compensate for changes in chemical stimuli (P_{aCO_2}). For given respiratory system mechanics, V_T /PTP-P_{mus} is heavily dependent on the mode of support. Thus, the effectiveness of chemical feedback in compensating for changes in chemical stimuli should be mode-

dependent. Modes of support that permit the intensity of patient inspiratory effort to be expressed on ventilator-delivered volume improve the effectiveness of chemical feedback in regulating PaCO_2 and particularly in preventing respiratory alkalosis. In normal conscious subjects receiving maximum assistance on the three main ventilator modes,¹⁶ the ability of the subject to regulate PaCO_2 depends on the operational principles of each mode, specifically in terms of $V_T/\text{PTP-Pmus}_I$ (Fig. 35-6). At all levels of CO_2 stimulation, preservation of neuroventilatory coupling increased progressively from ACV to PSV to PAV; the ability of subjects to regulate PaCO_2 followed the same pattern.¹⁶



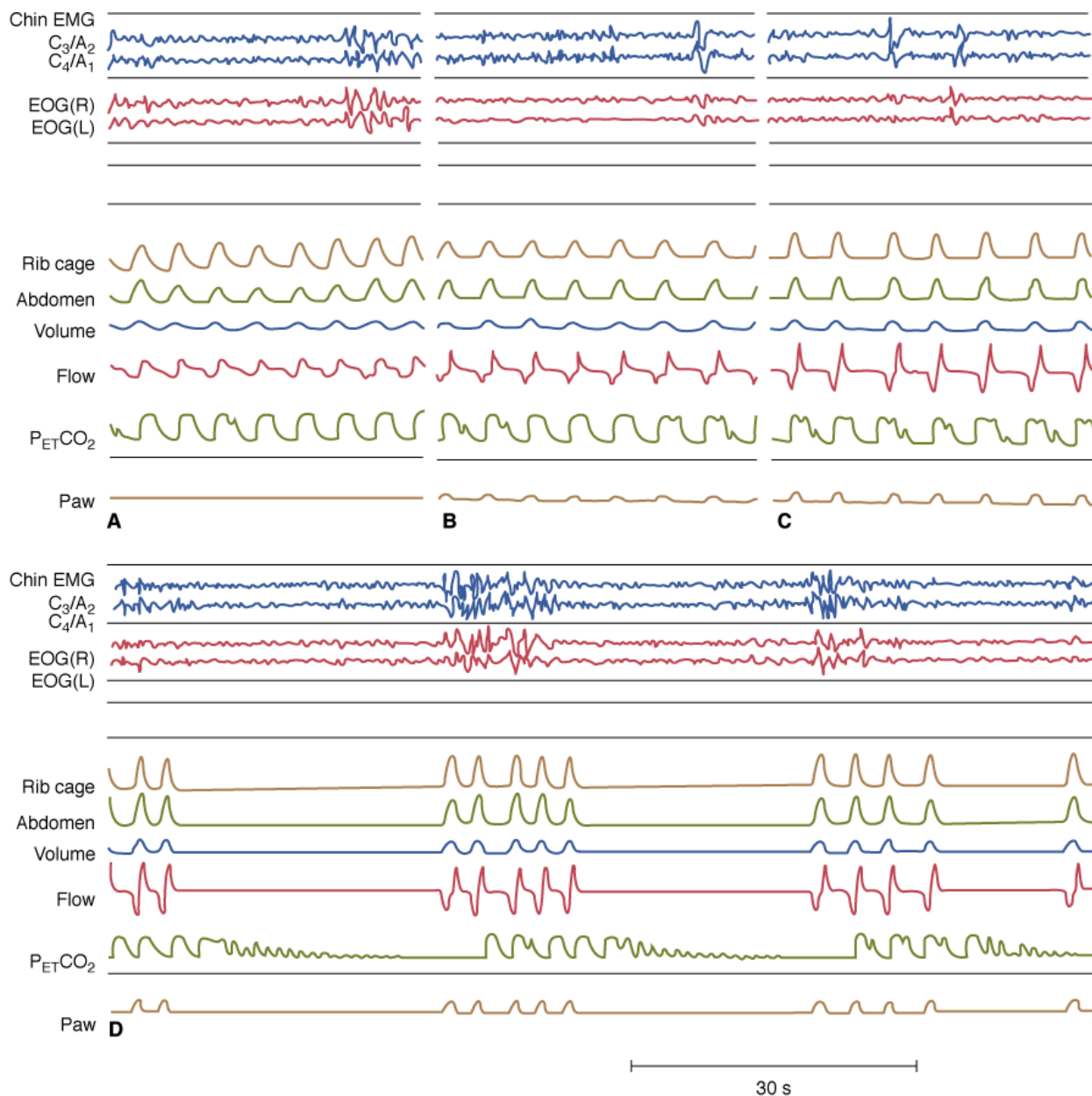
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Ratio (mean $\pm$ SD) of tidal volume to pressure-time product of inspiratory muscles ($V_T/\text{PTP-Pmus}_I$) in normal, conscious subjects ventilated with three modes of assisted ventilation in the absence and presence of CO_2 challenge (inspired CO_2 concentration increased in small steps until intolerance developed). *Open* and *closed bars* represent zero and final (highest) concentration of inspired CO_2 , respectively. *AVC*, assist-volume control; *PAV*, proportional-assist ventilation; *PS*, pressure-support ventilation. *Asterisk* indicates significant difference from the value without CO_2 challenge. *Plus* sign indicates significant difference from the corresponding value with PAV. With each mode, subjects were ventilated at the highest comfortable level of assistance (corresponding to 80% reduction of patient resistance and elastance with PAV, 10 cm H_2O of pressure support, and 1.2-L tidal volume with AVC). With CO_2 challenge, $V_T/\text{PTP-Pmus}_I$ decreased significantly when the subjects were ventilated with PS and AVC, but it remained relatively constant with PAV. Without CO_2 challenge, $V_T/\text{PTP-Pmus}_I$ was significantly higher with PS and AVC than with PAV. This response pattern caused severe respiratory alkalosis with PS and AVC (P_{ETCO_2} decreased to approximately 22 mm Hg with both modes) but not with PAV (P_{ETCO_2} approximately 30 mm Hg). Unlike with PS and PAV, subjects ventilated with AVC could not tolerate high values of P_{ETCO_2} (final P_{ETCO_2} was approximately 7, 11, and 13 mm Hg higher than baseline eupnea, respectively, with AVC, PS, and PAV). (Based on data from Mitrouska et al.¹⁶)

Neurally adjusted ventilatory assist (NAVA) is a new mode of support that, similar to PAV, uses patient effort to drive the ventilator.⁴⁵⁻⁴⁷ The electrical activity of the diaphragm is obtained with a special designed esophageal catheter and serves as a signal to link inspiratory effort to ventilator pressure (see Chapter 13). Because neuroventilatory coupling is preserved, the principles described above also apply to NAVA.^{46,47}

During sleep or sedation, the tendency to develop hypocapnia with ACV and PSV (see Chapter 57 for the effects of mechanical ventilation on sleep) may have serious consequences because a drop of a few millimeters of mercury in PaCO_2 leads to apnea and periodic breathing.^{8,19} Thus, excessive assistance with ACV and PSV promotes unstable breathing secondary to impaired neuroventilatory coupling; controller gain remains high in the face of low inspiratory effort (Fig. 35-7). Unstable breathing, however, during sleep secondary to mechanical ventilation may be prevented or attenuated with PAV and NAVA that does not guarantee a minimum V_T .^{8,19,46,47} Modes that decrease the volume delivered by a ventilator in response to any reduction in the intensity of patient effort enhance breathing stability and may be associated with better sleep quality.⁴⁸ Nevertheless, if the assist setting during PAV or NAVA is such that controller gain increases considerably, and the inherent loop gain of the patient is relatively high, the patient will be at risk of developing unstable breathing.^{23,33,41,49,50}

Figure 35-7



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Polygraph tracings in a healthy subject during non-rapid eye movement sleep with and without pressure-support ventilation. (A) Spontaneous breathing with continuous positive airway pressure (CPAP). (B) to (D) Pressure support of 3, 6, and 8 cm H₂O, respectively. Periodic breathing with central apneas developed with pressure support of 8 cm H₂O. C3/A2 and C4/A1, electroencephalogram channels; EMG, electromyogram; EOG, electrooculogram (right [R] and left [L]); Paw, airway pressure; PETCO₂, end-tidal P_{CO2}. (Used, with permission, from Meza, et al. Susceptibility to periodic breathing with assisted ventilation during sleep in normal subjects. *J Appl Physiol*. 2003;167:1193–1199.)

These principles may be altered by disease states and therapeutic interventions. Although little is known about the interaction between disease states and mechanical ventilation on control of breathing, two examples help in illustrating the point. First, in conscious patients with sleep apnea syndrome, a drop in PaCO₂ because of brief (40 seconds) hypoxic hyperventilation resulted, contrary to healthy subjects, in significant hypoventilation and triggering of periodic breathing in some patients.⁵¹ This hypoventilation was interpreted as evidence of a defect (or reduced effectiveness) of short-term poststimulus potentiation, a brainstem mechanism that promotes ventilatory stability.⁵¹ In this situation, a level of assistance that causes a significant decrease in PaCO₂ may promote unstable breathing in awake patients with sleep apnea syndrome, a situation closely resembling that observed during sleep. Second, studies in ventilated critically ill patients have shown that when awake patients are unable to increase V_T appropriately as a result of the mode used (i.e., PSV), they increase respiratory rate in response to a chemical challenge.⁵² Behavioral feedback, however, may underlie this response pattern. In sedated patients with acute respiratory distress syndrome (in whom behavioral feedback is not an issue) receiving PSV, considerable variation in PaCO₂ elicited a steady-state response limited to the intensity of breathing effort, a response pattern similar to that observed in normal subjects.^{9,16}

Neuromechanical Feedback

Intrinsic Properties of Respiratory Muscles

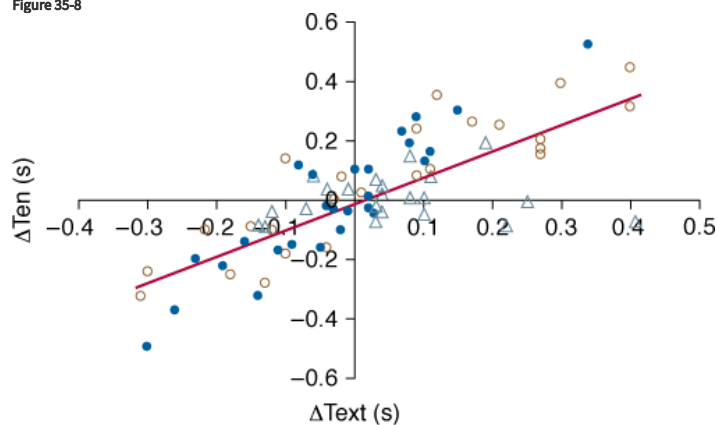
For a given neural output, P_{mus} decreases with increasing lung volume and flow, as dictated by the force-length and force-velocity relationships of inspiratory muscles, respectively.⁵³ Therefore, for a given level of muscle activation, P_{mus} should be smaller during mechanical ventilation than during spontaneous breathing if pressure provided by the ventilator results in greater flow and volume. It has been shown in healthy subjects ventilated with PSV that, compared with spontaneous breathing, the relationship between electrical activity (Edi) and pressure-time product of diaphragm (PTPdi) is shifted to the left; thus, at any given level of Edi , PTPdi is reduced.⁵⁴

The influence and consequences of mechanical feedback during mechanical ventilation have not been studied satisfactorily. It is possible that this type of feedback is of clinical significance in patients with dynamic hyperinflation (high end-expiratory lung volume), high ventilatory requirements (requirements for high flow and volume), and/or impaired neuromuscular capacity.

Reflex Feedback

The characteristics of each breath are influenced by various reflexes that are related to lung volume or flow and mediated, after a latency of a few milliseconds, by receptors located in the respiratory tract, lung, and chest wall.^{5,6} Mechanical ventilation may stimulate these receptors by changing flow and volume. In addition, changes in ventilator settings, inevitably associated with changes in volume and flow, also may elicit acute P_{mus} responses mediated by reflex feedback. In sedated patients with acute respiratory distress syndrome, manipulation of ventilator settings altered immediately (within one breath) the neural respiratory timing, whereas respiratory drive remained constant.^{9,55} Specifically, decreases in V_T and pressure support and increases in inspiratory flow caused an increase in respiratory frequency. Depending on the type of alteration, changes in respiratory frequency were mediated via alteration in neural inspiratory and expiratory time; increases in inspiratory flow caused increases in respiratory frequency mainly by decreasing neural inspiratory time; decreases in V_T and pressure support caused increases in respiratory frequency by decreasing neural expiratory time. This reflex response was similar, at least qualitatively, to that observed in healthy subjects during wakefulness and sleep.^{56–60} There was a strong dependency of neural expiratory time on the time that mechanical inflation extended into neural expiration; neural expiratory time increased proportionally to the increase in the delay between the ventilator cycling off and the end of neural inspiratory time (Fig. 35-8).^{9,55} This finding indicates that expiratory asynchrony may elicit a reflex timing response. A subsequent study in a general intensive care unit population confirmed the dependency of neural expiratory time on expiratory asynchrony.⁶¹ The most likely explanation for the timing response is the Herring-Breuer reflex.

Figure 35-8



$$y = -0.004 + 0.897x, P < 0.001$$

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Relationship between the changes in the time that mechanical inspiration extended into neural expiration ($\Delta Text$, expiratory asynchrony) and neural expiratory time (ΔTen) in mechanically ventilated patients with acute respiratory distress syndrome. Closed circles, open circles, and open triangles represent $\Delta Text$ induced by changes in volume (at constant flow), flow (at constant volume), and pressure support, respectively. Solid line, regression line. (Based on data from Kondili E, Prinianakis G, Anastasaki M, Georgopoulos D. Acute effects of ventilator settings on respiratory motor output in patients with acute lung injury. *Intensive Care Med*. 2001;27:1147–1157.)

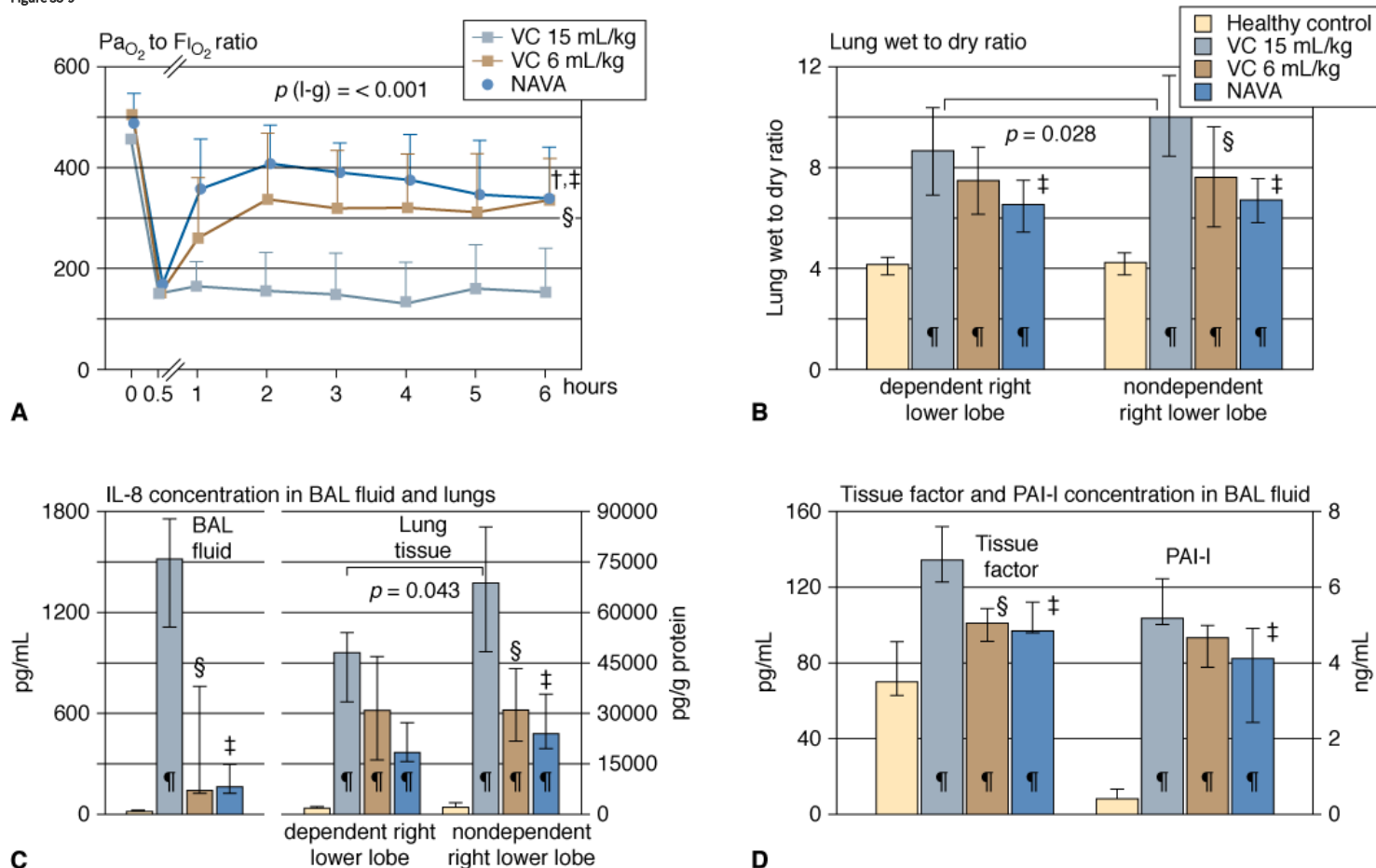
The final response may be unpredictable depending on the magnitude and type of lung volume change, the level of consciousness, and the relative strength of the reflexes involved. Nevertheless, reflex feedback should be taken into account when ventilator strategies are planned. A few examples may help in illustrating the importance of reflex feedback in patient-ventilator interaction. Assume that the patient is receiving pressure support that is being decreased during weaning. This results in lower V_T , which through reflex feedback decreases neural expiratory time, causing an increase in respiratory frequency.^{9,55} This increase should not be interpreted as patient intolerance to the decrease in pressure support. Consider another patient with obstructive lung disease

receiving ACV. V_T is decreased at a constant inspiratory flow so as to reduce the magnitude of dynamic hyperinflation (less volume is exhaled over a longer period). The lower V_T usually results in less delay in breath termination as compared with the end of neural inspiration, which through vagal feedback will decrease neural expiratory time, limiting the effectiveness of this strategy for reducing dynamic hyperinflation.⁵⁵ Assume in another patient receiving ACV that inspiratory flow is increased at a constant V_T , with the intent of reducing inflation time and providing more time for expiration so as to reduce dynamic hyperinflation. This step causes a reflex decrease in neural inspiratory time and an increase in respiratory frequency. Mechanical expiratory time may change in either direction depending mainly on the relation between neural and mechanical inspiratory time. In patients receiving ACV, expiratory time showed a variable response to changes in flow rate; some patients actually demonstrate a reduced expiratory time with a higher flow,⁶² which cancels the desired reduction in dynamic hyperinflation.

There are neural reflexes that inhibit inspiratory muscle activity if lung distension exceeds a certain threshold, which is well below total lung capacity (Hering-Breuer reflex).^{6,63,64} These reflexes protect the lung from overdistension, which is associated with lung injury.^{65,66} Pressure-control or volume-control modes of assisted ventilation considerable interfere with the ability of these reflexes to regulate tidal volume.^{16,67} With these modes, as a result of neuroventilatory uncoupling (high $V_T/PTP-P_{mus}$), overassistance may result in high tidal volume leading to regional or global lung overdistension. Conversely, recent evidence indicates that ventilator modes that permit reflex feedback to regulate the tidal volume and respiratory rate (viz., NAVA, PAV) may protect against or lessen ventilator-induced lung injury.

Brander et al⁶⁸ randomized anesthetized rabbits with early experimental acute lung injury into three ventilator strategies: NAVA (nonparalyzed), volume control with tidal volume of 6 mL/kg (paralyzed, protective strategy), and volume control with tidal volume of 15 mL/kg (paralyzed, injurious strategy). Animals randomized to NAVA selected an average tidal volume of 2.7 ± 0.9 mL/kg and respiratory rate up to three times higher than that in both controlled ventilation groups—a breathing-pattern response that can be explained by vagally controlled reflexes.^{6,63,64} Compared to the 15 mL/kg group, animals ventilated with either NAVA or volume control at 6 mL/kg exhibited less ventilator-induced lung injury, as indicated by lung injury scores, lung wet-to-dry ratio, and lung and systemic biomarkers (Fig. 35-9). These results indicate that the use of NAVA, which allowed the animals to choose their own respiratory pattern, was at least as effective in preventing various manifestations of ventilator-induced lung injury as conventional, volume-controlled ventilation using a tidal volume of 6 mL/kg.

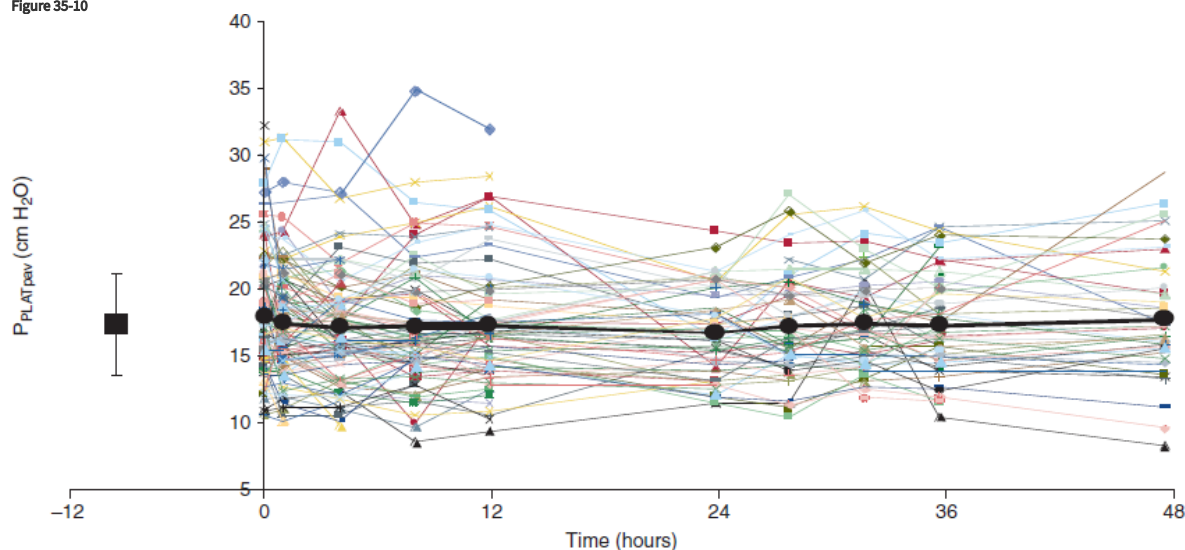
Figure 35-9



Parameters indicative of ventilator-induced lung injury (VILI) in rabbits with induced acute lung injury (ALI) and ventilated with three strategies: NAVA, volume control with tidal volume (V_T) of 6 mL/kg, and volume control with V_T of 15 mL/kg. **(A)** There were no differences in partial pressure of arterial oxygen to fractional inspired oxygen concentration ratio (P_{aO_2}/F_{IO_2}) among groups before and 30 minutes after induction of ALI. The increase in P_{aO_2}/F_{IO_2} shortly after switching to the assigned ventilation mode (i.e., after randomization into the treatment groups) was more pronounced with NAVA than with volume control (VC) 6-mL/kg ($p < 0.05$ post hoc analysis), although P_{aO_2}/F_{IO_2} was not different between NAVA and VC 6-mL/kg at the end of the protocol. With VC 15-mL/kg, P_{aO_2}/F_{IO_2} remained below 200. **(B)** The lung wet-to-dry ratio with NAVA and with VC 6-mL/kg was lower than with VC 15-mL/kg (albeit not significantly for the dependent lung in VC 6-mL/kg animals). **(C)** and **(D)** Interleukin 8 (IL-8), tissue factor, and plasminogen activator inhibitor type 1 (PAI-1) concentration in bronchoalveolar (BAL) fluid was higher in all study groups compared to healthy controls and was higher with VC 15-mL/kg than with the other two groups (except for PAI-1 in VC 6-mL/kg). Lung tissue IL-8 concentration was increased in all groups as compared to nonventilated controls and was highest in the nondependent lung regions with VC 15-mL/kg. In the VC 6-mL/kg and NAVA groups, lung tissue IL-8 concentration was lower compared to VC 15-mL/kg (albeit not significant for the dependent lung region). Groups are shown as mean $\pm$ standard deviation (SD) for A and B, or as median (quartiles) for C and D. Symbols represent group mean; bars indicate standard deviation. e–g, time–group interaction (two-way analysis of variance). Post hoc pairwise comparison procedure between groups: $\dagger p < 0.05$ NAVA versus VC 6-mL/kg; $\ddagger p < 0.05$ NAVA versus VC 15-mL/kg; $\S p < 0.05$ VC 6-mL/kg versus VC 15-mL/kg. (Used, with permission, from Brander L, Sinderby C, Lecomte F, et al. Neurally adjusted ventilatory assist decreases ventilator-induced lung injury and non-pulmonary organ dysfunction in rabbits with acute lung injury. *Intensive Care Med.* 2009;35:1979–1989.)

In a human study employing randomized design, Xirouchaki et al.⁶⁹ ventilated 108 critically ill patients, most of whom had acute lung injury or acute respiratory distress syndrome, with PAV+ (PAV with automatic estimation of elastance and resistance of the respiratory system; see Chapter 12). Even with high assistance, tidal volume and end-inspiratory plateau pressure were comparable to those observed during protective controlled mechanical ventilation. Examination of individual end-inspiratory plateau pressures during PAV+ showed that out of a total of 744 measurements only on nine occasions (1.2%) and in five patients (4.6%) were plateau pressures above 30 cm H₂O (Fig. 35-10). Ninety-four percent of the end-inspiratory plateau pressures were below 26 cm H₂O, a value associated with lung protection.⁷⁰ Similar to the findings of Brander et al, these results can be explained by the operation of reflex feedback (vagal controlled reflexes).^{6,63,64}

Figure 35-10



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Individual values of quasi-static airway pressure obtained by 300 msec pause maneuver at the end of selected inspirations ($P_{PLAT_{pav}}$) as a function of time in 108 critically ill patients randomized (zero time) to proportional assist ventilation with load-adjustable gain factors (PAV+). PAV+ was continued for 48 hours unless the patients met predefined criteria, either for switching to controlled modes or for breathing without ventilator assistance. Closed black circles connected by solid thick line represent mean values. Each patient is denoted by a single color. For comparison the mean $\pm$ standard deviation (SD) values of static end-inspiratory airway pressure, obtained within 8 hours before randomization during controlled mechanical ventilation (CMV), is shown (closed black square). Notice that in the majority of the patients $P_{PLAT_{pav}}$ was below 26 cm H₂O. (Used, with permission, from Kondili et al. Patient-ventilator interaction. *Br J Anaesth.* 2003;91:106–119.)

Is it possible to use this reflex feedback into a clinical scenario? Recent studies suggest that in patients with acute respiratory distress syndrome titration of tidal volume based on individual lung mechanics may be a better strategy than using a fixed tidal volume (i.e., 6 mL/kg).^{65,66,70–72} Obtaining lung mechanics, however, necessitates the use of cumbersome techniques not easily available at bedside. Theoretically, tidal volume selected by the patient should be based on individual lung mechanics, which serve as a guide for setting the ventilator.^{73,74} Although studies support this hypothesis,^{68,69} caution

should be exercised in patients with strong signals of nonrespiratory origin (acidosis, brain dysfunction) that drive ventilation. Notwithstanding the limitations and feasibility of this approach, this hypothesis deserves further studies.

Neuromechanical Inhibition

Mechanical ventilation at relatively high tidal volume and ventilator frequency results in a non-chemically mediated decrease in respiratory motor output.^{75–77} This decrease, referred to as *neuromechanical inhibition*, is manifested both in respiratory frequency and in amplitude of respiratory motor output. Neuromechanical inhibition lasts for several breaths after termination of mechanical ventilation, thus constituting a type of control system inertia and resetting of the spontaneous respiratory rhythm.⁷⁸ Although the mechanism underlying neuromechanical inhibition is not entirely clear, the Hering-Breuer reflex is the most plausible explanation. In addition, Sharshar et al⁷⁹ showed that mechanical ventilation reduces the excitability of cortical motor areas representing respiratory muscles. It is possible that mechanoreceptor feedback accounts for the depression of the motor-evoked potential of the diaphragm via vagal and other proprioceptive afferents to the respiratory center. The clinical relevance of neuromechanical inhibition is currently unknown. Available evidence suggests that its contribution to respiratory motor output in ventilated critically ill patients is rather minimal.^{9,11,55}

Entrainment of Respiratory Rhythm to Ventilator Rate

Entrainment of respiratory rhythm to the ventilator rate implies a fixed, repetitive, temporal relationship between the onset of respiratory muscle contraction and the onset of a mechanical breath.^{80–82} Human subjects exhibit one-to-one entrainment over a considerable range above and below the spontaneous breathing frequency.^{83,84} Cortical influences (learning or adaptation response) and the Hering-Breuer reflex are postulated as the predominant mechanisms of entrainment. Theoretically, one-to-one entrainment should facilitate patient-ventilator synchrony, but studies of the entrainment response in critically ill patients are lacking.

Behavioral Feedback

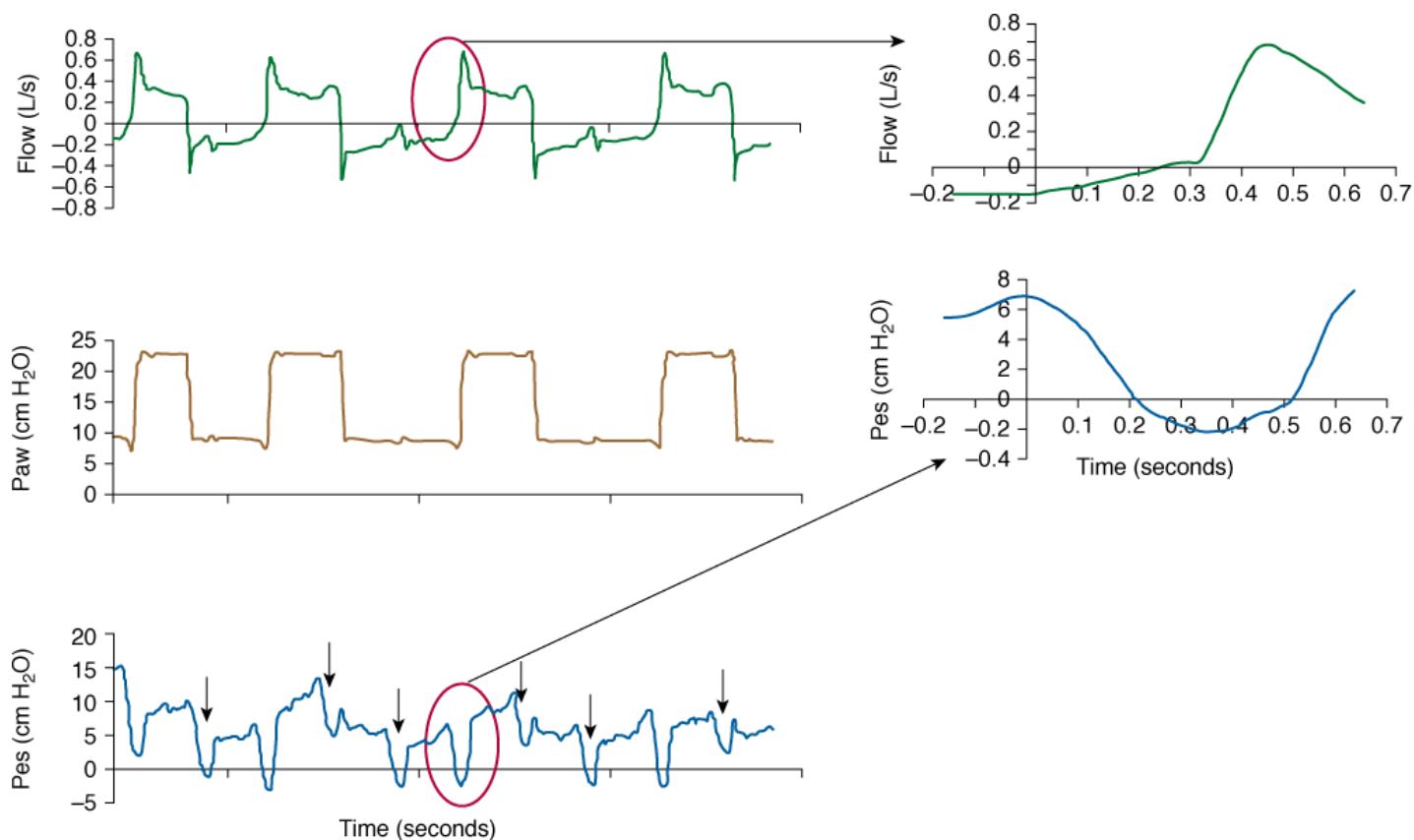
The effects of behavioral feedback on control of breathing in ventilated patients are unpredictable, depending on several factors related to the individual patient and surroundings. Alteration in ventilator settings, planned to achieve a particular goal, might be ineffective in awake patients because of behavioral feedback.^{85,86} Inappropriate ventilator settings may cause breathing discomfort in awake patients. Consequent panic reactions further aggravate the unpleasant breathing sensation and create a vicious cycle. Behavioral feedback also may be altered considerably from time to time secondary to changes in the level of sedation, sleep-awake state, patient status, and environmental stimuli. The many factors involved in behavioral feedback complicate its study and the interpretation of its effects on the system that controls breathing in mechanically ventilated patients.

Interactive Effects of Patient-Related Factors and Ventilator on Control of Breathing

Mechanics of Respiratory System

The mechanical properties of the respiratory system may influence the pressure delivered by the ventilator independent of patient effort and thus may modify the effects of mechanical ventilation on the various feedback loops. Excessive triggering delay and ineffective triggering are common in patients with obstructive lung disease and dynamic hyperinflation (Fig. 35-11). In the setting of airflow obstruction, mathematical models predict that PSV can be accompanied by marked variation in V_T and intrinsic positive end-expiratory pressure even when patient effort is constant.⁸⁷ This dynamic instability increases as the time constant of the respiratory system increases and produces patient-ventilator asynchrony of variable magnitude and type. The demonstration of increased arousals during PSV, but not during volume-cycled ventilation, may be caused in part by dynamic patient-ventilator asynchrony.²¹

Figure 35-11



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Airflow (inspiration up), airway pressure (P_{aw}), and esophageal pressure (P_{es}) in a patient with obstructive lung disease ventilated with pressure support. Note the triggering delay with every mechanical breath (see the magnified tracing of flow and P_{es}) and the ineffective efforts (arrows). The ventilator rate was 12 breaths/min, whereas the patient's respiratory frequency was 35 breaths/min. Extrapolation from ventilator rate to the patient's system of control of breathing is misleading (Used, with permission, from Springer Science and Business Media: Brander, et al. *Intensive Care Med*. 2009;35:1979–1989.)

Ineffective triggering has been observed with all modes of assisted ventilation. It is particularly common with tachypnea and when the level of assistance is relatively high and mechanical inflation extends well into neural expiration.^{11,67,88,89} With PAV and NAVA, the likelihood of ineffective efforts is reduced significantly because mechanical inflation time is terminated close to the end of neural inspiration, and tidal volume in most cases remains relatively small.^{46,47,67,69}

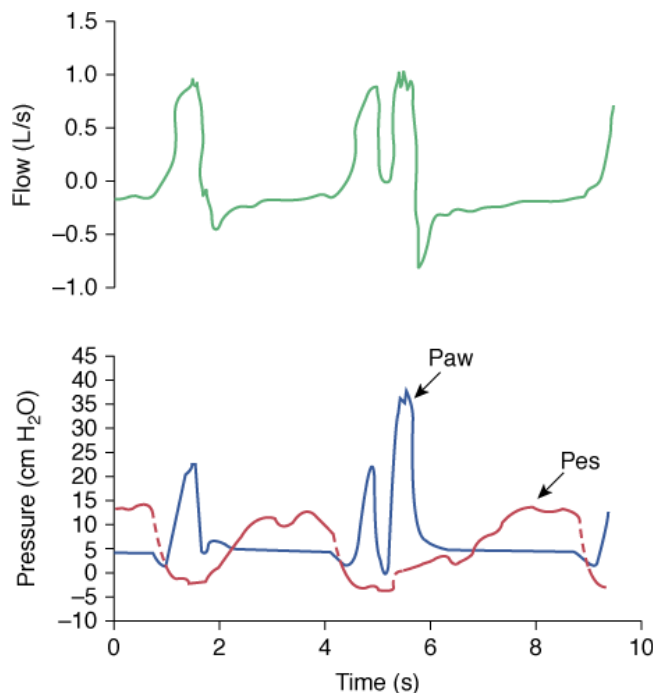
The phenomenon of ineffective efforts considerably influences the interpretation of ventilatory output in relation to the control of breathing during mechanical ventilation.^{3,4} In the presence of ineffective efforts, ventilator frequency does not reflect the patient's spontaneous respiratory rate (see Fig. 35-11). Moreover, with ineffective efforts, significant alteration in a patient's respiratory effort occurs secondary to changes in feedback loop.

Characteristics of the Muscle Pressure Waveform

The characteristics of the P_{mus} waveform influence the ventilator-delivered volume in a complex manner, depending on several patient and ventilator factors. Extensive review of these factors is beyond the scope of this chapter, but some examples are provided.

The initial rate of increase in P_{mus} interacts with triggering of the ventilator.¹¹ A low rate of initial increase in P_{mus} occurs with a concave upward shape of P_{mus} or a low respiratory drive (such as with low P_{aCO_2} , sedation, sleep, or a high level of assistance); this increases the time delay between the onset of patient inspiratory effort and ventilator triggering and promotes asynchrony. In the presence of dynamic hyperinflation, a prolonged triggering time, particularly when associated with a relatively short neural inspiratory time and low peak P_{mus} , may result in ineffective efforts. Alternatively, an increase in the intensity of inspiratory effort, such as occurs with an increase in metabolic rate, high P_{aCO_2} , or decrease in the level of sedation or assistance, is manifested both in the rate of rise and in the peak of P_{mus} . The change may cause a decrease in the time delay, thus promoting patient-ventilator synchrony.¹¹ If, however, patient inspiratory effort is vigorous and longer than mechanical inflation time, the ventilator may be triggered more than once during the same inspiratory effort (Fig. 35-12).^{3,90} It follows that changes in the characteristics of the P_{mus} waveform may influence the ventilator rate and ventilatory output despite no change in a patient's breathing frequency. Alterations in ventilatory output may secondarily modify patient effort through changes in feedback loops (see Fig. 35-1).

Figure 35-12



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Flow, airway pressure (Paw), and esophageal pressure (Pes) in a patient recovering from acute lung injury and ventilated on assist volume control at constant inspiratory flow. In the second breath, tidal volume (volume was not shown) was decreased at constant inspiratory flow. As a result, there was premature termination of mechanical inspiration. Because the inspiratory muscles continued to contract, they developed sufficient pressure to overcome elastic recoil at end inspiration. As a result, Paw decreased below the triggering threshold, and the ventilator delivered a new mechanical breath. The ventilator was triggered three times by the two inspiratory efforts. Note the high Paw of the third mechanical breath secondary to high lung volume (the volume of the third breath was added to that of the second). Total breath duration of the second respiratory effort was considerably longer than that of the first effort owing to activation of Hering-Breuer reflex by the high volume. (Used, with permission, from Springer Science and Business Media: Xirouchaki, et al. *Intensive Care Med*. 2008;34:2026–2034.)

The Future

Over the past two decades, many studies have been performed in animals and human subjects with an aim of improving a patient's ability to control the ventilator. Various ventilator modes target either an improvement in the response of the ventilator to patient effort or tight coupling between the ventilator-delivered pressure and patient instantaneous ventilatory demands. Studies of these modes have yielded promising results. New methods of triggering have been shown to improve the response of the ventilator to patient effort.^{45,91–93} Algorithms that automatically adjust the criterion for cycling off have been designed with a goal of reducing expiratory asynchrony.⁹⁴ Estimates of the inspiratory muscle pressure waveform may also be used to terminate pressure delivery, and these, theoretically, should improve patient-ventilator synchrony.⁹³ Mechanical⁹² and electrical^{46,95} activity of the diaphragm has been used to control the level and duration of inspiratory assistance. With PAV, methods of noninvasive automatic estimation of elastance and resistance of the respiratory system are now available (PAV+),^{96,97} which enable controller gain to be maintained constant in the face of changes in the mechanical load of respiratory system⁹⁸ and result in fewer intervention in terms of ventilator settings compared to other modes.⁹⁹ Algorithms that use a signal generated from flow, volume, and airway pressure may be used to provide breath-by-breath quantitative information of inspiratory muscle pressure,¹⁰⁰ and this approach also may be used in the future to facilitate patient-ventilator synchrony. By achieving tight coupling between neural output and ventilator-delivered pressure, the ventilator is able to serve as a respiratory muscle with high capabilities and operate in harmony with the system that controls breathing. Nowadays, it seems feasible to shift from the physician who dictates the pattern of ventilation to the patient who chooses to breathe with a pattern that incorporates all the aspects of control of breathing. Because the control of ventilation is much more complex than simply regulating blood gases, it is likely that the patient can do a better job than physicians can.

Negative-feedback methods, such as adaptive pressure-support servoventilation, have been designed recently with a goal of reducing periodic breathing through appropriate changes in the level of assistance and maintaining a target minute ventilation in the face of waxing and waning respiratory efforts.^{82,101} Incorporation of this approach in assisted modes may decrease the propensity of high-risk individuals to develop periodic breathing. It is not known whether this mode could decrease morbidity in critically ill patients, although it should enhance sleep efficiency.^{22,23}

Conclusion

Incorporating an auxiliary pressure into the system that controls breathing changes the volume–time profile of a breath. It also alters, via chemical, neuromechanical, and behavioral feedback, the pressure developed by the respiratory muscles. The latter, depending on ventilator and patient factors, may or may not modify the auxiliary pressure. The response of patient effort to a ventilator-delivered breath and the response of a ventilator to patient effort are the two essential components of control of breathing during mechanical ventilation. The physician dealing with a ventilated patient should be aware that both the basic features of control of breathing and its expression can be altered considerably by the process of mechanical ventilation.

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